

Siyuan (Wilson) Sun

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EDUCATION

University of Arizona

Tucson, AZ

B.S. in Artificial Intelligence; GPA: 4.0/4.0

Aug. 2025 – May 2027 (Expected)

- **Relevant Coursework:** Deep Learning for NLP, Neural Networks, Reinforcement Learning, Principles of ML, Text Retrieval and Web Search.

East China University of Science and Technology

Shanghai, China

B.S. in Artificial Intelligence (Transferred to UArizona)

Sep. 2023 – July 2025

RESEARCH EXPERIENCE

LUMIA Lab, Shanghai Jiao Tong University

Shanghai, China

Research Intern (Advised by Prof. Zhouhan Lin)

Apr. 2026 – Present

- Researching **parametric memory for LLMs**, LLM pretraining, and test-time training (TTT).

CLULab, University of Arizona

Tucson, AZ

Undergraduate Research Assistant (Advised by Prof. Mihai Surdeanu)

Sep. 2025 – Present

- Researching **LLMs and Information Retrieval**, with a focus on Transformer-based neural retrieval systems and query expansion integration.

TECHNICAL SKILLS

Languages & Tools: Python, C/C++, Java, Shell, Linux, Git

Frameworks: PyTorch; Hugging Face Transformers, Accelerate, PEFT, TRL

Retrieval & Training: FAISS, Elasticsearch, Milvus; Multi-GPU Training, Mixed Precision (FP16/BF16)

SELECTED RESEARCH & PROJECTS

Deep Fusion Memory (DFM) | LLMs, Parametric Memory

ICLR 2027 (Under Review)

- *Deep Fusion Memory: A Chunk-Level Parametric Memory for Large Language Models.* Learns chunk-level memory and fuses it into a frozen LLM via gated cross-attention; no explicit retrieval at inference.
- Improved average scores over the strongest baseline by **12.71%** / **11.19%** on six QA tasks (Mistral-7B / Qwen3-8B); domain adaptation gains of **6.51** / **4.38 points** on Qwen3-4B / 8B.

AnchorQE | Dense Retrieval, Query Expansion

WSDM 2027 (Under Review)

- *Query Expansion Is More Than Generation: Improving Dense Retrieval through Better Integration.* Siyuan Sun and Mihai Surdeanu, 2026. Separately encodes queries and expansions, then interpolates their vectors using an unsupervised online weight estimate.
- Improved retrieval effectiveness by up to **12.89%** across TREC-DL, LoTTE, and BEIR without task-specific training; online calibration beat development-tuned fixed weights by up to **3.81%**.

Emotion Classification in Creative Text | RoBERTa, Domain Adaptation

- Transferred emotional priors from social media to poetry/lyrics with two-stage training; compared four tuning strategies, with full fine-tuning achieving the best poetry $F1 \approx 0.60$.

Learning Dense Reward Functions in Chess | Gymnasium, ResNet, IRL

- Inferred dense rewards from expert chess demonstrations with a custom Gymnasium environment and ResNet reward model; saliency analysis revealed human-like positional patterns.

Brain-to-Text: Decoding Intracortical Speech | BERT, GPT-2, LoRA

Brain-to-Text '25

- Built a BERT-GPT-2 neural speech decoder using masked phoneme modeling and LoRA, achieving **12.4% WER**; a day-specific adaptation layer handled non-stationary neural signals.